

Projectile Motion — Problem Set

Warm-up

1. A particle is projected with an initial velocity vector $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j}$. Find the initial speed and the angle of projection to the nearest degree.
2. Given the acceleration vector $\mathbf{a} = -10\mathbf{j}$, find the velocity vector $\mathbf{v}$ at time t if the initial velocity is $20\mathbf{i} + 15\mathbf{j}$.
3. A ball is kicked from the origin with $\mathbf{r}(t) = (15t)\mathbf{i} + (20t - 5t^2)\mathbf{j}$. Find the time it takes for the ball to return to the ground.
4. Using the displacement vector $\mathbf{r}(t) = (10t)\mathbf{i} + (12t - 5t^2)\mathbf{j}$, calculate the maximum height reached by the particle.
5. Find the horizontal range of a projectile whose displacement is given by $\mathbf{r}(t) = (8t)\mathbf{i} + (10t - 5t^2)\mathbf{j}$.
6. A projectile has a horizontal velocity component $\dot{x} = 10\sqrt{3}$ and a vertical component $\dot{y} = 10$. Find the magnitude and direction of the velocity.
7. If $y = 20t - 5t^2$, show that the time taken to reach maximum height is $t = 2$.
8. Convert the parametric equations $x = 12t$ and $y = 16t - 5t^2$ into a Cartesian equation for the path of the projectile.
9. A stone is dropped from a height of 45m. Using $\ddot{y} = -10$, find the time it takes to hit the ground.
10. A particle is projected at 30° to the horizontal with speed 40 m/s. Write the initial velocity in vector form $\dot{x}\mathbf{i} + \dot{y}\mathbf{j}$.

Skill-building

1. A projectile is launched from the origin with initial speed V at angle θ .
 - a) Derive the expression for the time of flight T in terms of V, θ and g .
 - b) Show that the range is given by $R = \frac{V^2 \sin 2\theta}{g}$.
2. A particle is projected from a point 20m above the ground with $\mathbf{v}(0) = 10\mathbf{i} + 20\mathbf{j}$.
 - a) Show that the vertical displacement is $y = -5t^2 + 20t + 20$.
 - b) Find the time taken to reach the ground.
3. A ball is thrown at an angle α such that $\tan \alpha = \frac{3}{4}$ with an initial speed of 25 m/s. Find the velocity vector after 2 seconds (Take $g = 10$).
4. Show that the Cartesian equation $y = x \tan \theta - \frac{gx^2}{2V^2 \cos^2 \theta}$ can be written as $y = x \tan \theta - \frac{gx^2}{2V^2} (1 + \tan^2 \theta)$.

5. A projectile reaches a maximum height of 20m and has a horizontal range of 80m. Find the angle of projection θ .
6. A particle is projected from the origin. Its position at time t is $\mathbf{r}(t) = (V \cos \theta)t\mathbf{i} + ((V \sin \theta)t - 4.9t^2)\mathbf{j}$.
 - a) Find the time $t > 0$ when the particle is at a height of 10m if $V = 40$ and $\theta = 30^\circ$.
 - b) Calculate the horizontal distance traveled at this time.
7. A stuntman is projected from a platform 10m high with speed 20 m/s at an angle of 45° .
 - a) Find the time of flight.
 - b) Find the speed at which he hits the ground.
8. For a projectile launched at angle θ with speed V , show that the speed v at any time t is given by $v = \sqrt{V^2 - 2Vgt \sin \theta + g^2 t^2}$.
9. Two particles are projected simultaneously from the same point with the same speed V but at angles θ and $(90^\circ - \theta)$. Show that they have the same horizontal range.
10. A target is located at (X, Y) . Show that if a projectile is to hit the target, the angle of projection θ must satisfy:

$$\frac{gX^2}{2V^2} \tan^2 \theta - X \tan \theta + \left(Y + \frac{gX^2}{2V^2} \right) = 0$$

Easier Exam Questions

1. (Gosford 2020 Q12d) A particle is projected across horizontal ground from the origin O . Its initial velocity vector is $12\mathbf{i} + 5\mathbf{j}$ and its acceleration vector is $0\mathbf{i} - 10\mathbf{j}$.

Find:

- (i) the initial speed of the particle 1
- (ii) the angle of projection, correct to the nearest minute 1
- (iii) Beginning with its acceleration vector, derive the velocity at any time t in the form

$$\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$$

- (iv) Show that its displacement at time t is $\mathbf{r} = (12t)\mathbf{i} + (5t - 5t^2)\mathbf{j}$ 1
- (v) Find the maximum height of the particle. 1

2. (Sydney Boys 2022 Q14a) A golfer hits a golf ball from a point O with speed V m/s at an angle of 30° above the horizontal and moves freely under gravity. The ball reaches its greatest height at time T seconds after projection.

The position vector, $\underline{r}$, at any time, t seconds, is given by

$$\underline{r}(t) = \left(\frac{\sqrt{3}}{2}Vt - 5t^2 \right) \mathbf{i} + \left(\frac{1}{2}Vt - 5t^2 \right) \mathbf{j} \quad (\text{Do NOT prove})$$

Find, in terms of V , the speed of the ball at time $\frac{2}{3}T$ seconds after projection.

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3. (Caringbah 2021 Q14d) When a particle is fired in the open from a point O at a speed of 40 ms^{-1} and at an angle of θ above the horizontal, where $0 < \theta < \frac{\pi}{2}$. You may assume without proof that the horizontal displacement (x metres) and the vertical displacement (y metres) of the particle from O at time t seconds after firing are given by $x = 40t \cos \theta$ and $y = 40t \sin \theta - 5t^2$. The particle is fired with the same speed from a point O on the floor of a horizontal tunnel that has a height of 20 metres. (You may assume the projectile has no size and can reach the maximum height of 20 metres without touching the roof of the tunnel)

- (i) Show that the time taken for the projectile to reach the maximum height is $t = 4 \sin \theta$ **1**
- (ii) Find the angle of projection for this maximum height in the tunnel. **2**
- (iii) Show that the exact maximum horizontal range for the projectile in this tunnel is $80\sqrt{3}$ metres. **2**

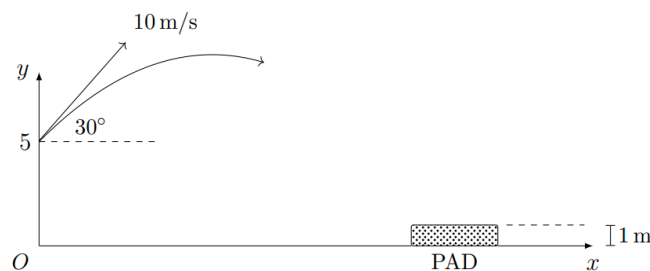
4. (Cherrybrook Tech 2021 Q14a) A particle is projected upwards from the top of a building 12 metres high with an initial velocity of 40 ms^{-1} . The particle is launched at an angle of 30 degrees with the horizontal.

Take 9.8 ms^{-2} for the acceleration due to gravity.

- (i) If the origin is at the base of the building, show that the displacement vector of the particle is: **3**

$$\underline{s}(t) = 20\sqrt{3} t \underline{i} + (-4.9t^2 + 20t + 12)\underline{j}$$

- (ii) Show that the time of flight is 4.61 seconds, correct to 2 decimal places. **1**
 - (iii) Determine the speed, correct to 2 decimal places, and the angle, correct to the nearest degree, at which the particle hits the ground. **3**
5. (Ascham 2024 Q12d) From a point five metres above level ground a stuntman is projected at an angle of elevation of 30° and at an initial speed of 10 m/s. A one metre high pad is placed on the ground to cushion his impact.



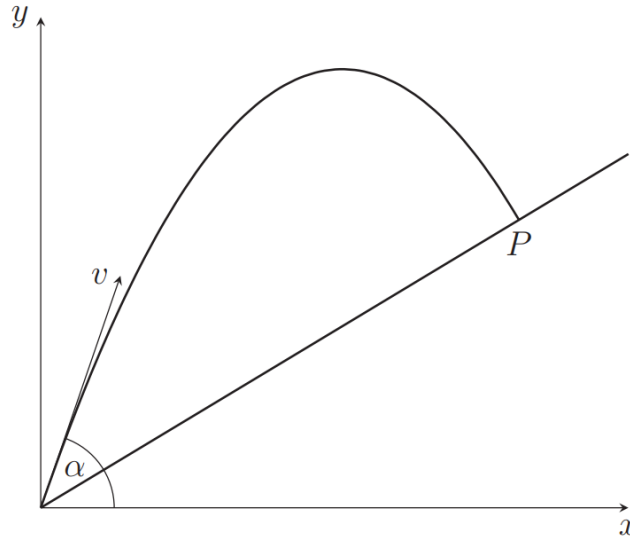
The displacement vector function is

$$\mathbf{r} = 5\sqrt{3}t\mathbf{i} + 5(1 + t - t^2)\mathbf{j}. \quad (\text{DO NOT SHOW THIS})$$

- (i) Show that the time of flight for the stuntman to land on the pad is approximately 1.52 seconds. **1**
- (ii) Determine the distance from O , correct to the nearest centimetre, where the pad should be centred, to hopefully cushion his impact. **1**
- (iii) Calculate his impact speed, to the nearest m/s. **2**
- (iv) This is to be performed at an indoor stadium with a ceiling height of 8 metres. Calculate the distance the stuntman will miss the ceiling by at his highest point. **2**

Harder Exam Questions

1. (Sydney Grammar 2025 Q14a) As part of avalanche control programs, small explosives are fired into mountainsides. An explosive is fired upwards along a mountainside that has a gradient of 0.6. The explosive is launched with an initial speed of $v \text{ ms}^{-1}$ at an angle of α to the horizontal, and hits the mountainside at the point of impact P , as shown in the diagram below.



The position of the projectile after t seconds is given by $\underline{r} = \begin{bmatrix} vt \cos \alpha \\ vt \sin \alpha - 5t^2 \end{bmatrix}$

Do NOT prove this.

- (i) Show that the Cartesian equation of the path of the projectile is given by $y = x \tan \alpha - \frac{5x^2}{v^2} \sec^2 \alpha$. **1**
- (ii) Show that the horizontal distance of P , in metres, from the point of projection is given by $\frac{1}{5}v^2 \cos^2 \alpha (\tan \alpha - 0.6)$. **2**
- (iii) Given that the path of the explosive is perpendicular to the mountainside at the point of impact, show that the angle of projection is approximately 71° , correct to the nearest degree. **2**
- (iv) Hence, determine the impact speed as a percentage of the initial speed. Give your answer correct to the nearest percent. **3**

2. (Fort Street 2022 Q13a) A particle, experiencing vertical acceleration due to gravity g and no air resistance, is projected over horizontal ground at speed $v \text{ ms}^{-1}$ at an acute angle θ to the horizontal. You may assume the six equations of motion listed below.

$$\ddot{x} = 0$$

$$\ddot{y} = -g\hat{j}$$

$$\dot{x} = v \cos \theta \hat{i}$$

$$\dot{y} = (-gt + v \sin \theta)\hat{j}$$

$$x = vt \cos \theta \hat{i}$$

$$y = \left(-\frac{g}{2}t^2 + vt \sin \theta\right)\hat{j}$$

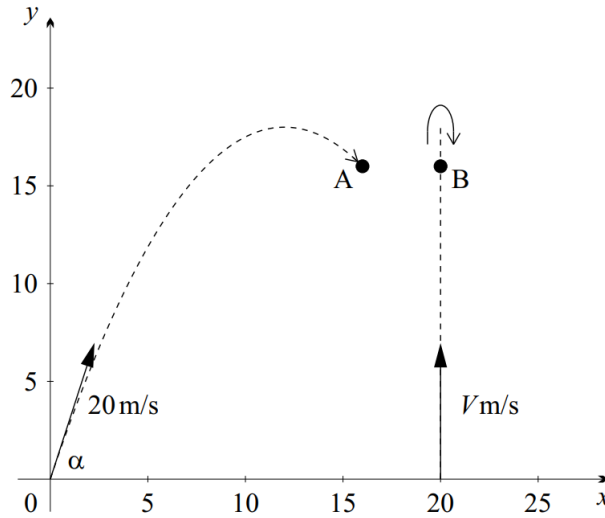
- i) Show that the

a) time to reach greatest height is $\frac{v \sin \theta}{g}$

1

- b) greatest height is $H = \frac{v^2 \sin^2 \theta}{2g}$ 1
- c) time to reach the landing point is $\frac{2v \sin \theta}{g}$ 1
- d) range is $R = \frac{v^2 \sin 2\theta}{g}$ 1
- ii) If R is three times H , find the exact angle of projection. 2

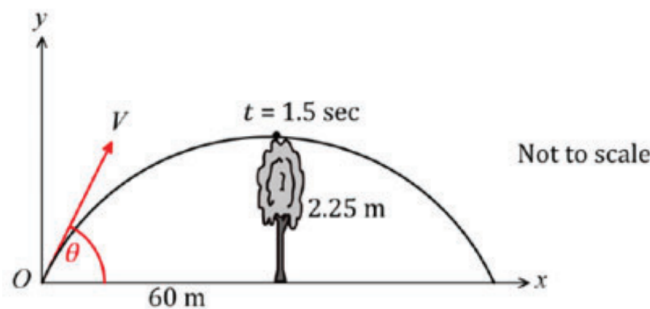
3. (Sydney Grammar 2023 Q13b) Two objects are launched at the same time from horizontal ground 20 m apart. Object A is projected from the origin at 20 m/s at an angle α degrees towards Object B and Object B is projected vertically at V m/s. The objects collide on their downward journey 10 m above the ground.



The positions $\tilde{r}_A$ and $\tilde{r}_B$ of Object A and Object B respectively after t seconds are given by the vector equations below (you do NOT need to prove these).

$$\tilde{r}_A = \begin{bmatrix} 20t \cos \alpha \\ -5t^2 + 20t \sin \alpha \end{bmatrix} \text{ and } \tilde{r}_B = \begin{bmatrix} 20 \\ -5t^2 + Vt \end{bmatrix}$$

- (i) Given that $45^\circ < \alpha < 90^\circ$, show that $\alpha \approx 72^\circ$. 3
- (ii) Hence find V . Write your answer correct to 2 decimal places. 2
4. (Girraween 2022 Q15b) A golfer hits a golf ball from a point O with speed V metres per second at an angle of θ above the horizontal, where $0 \leq \theta \leq \frac{\pi}{2}$. The ball just clears a 2.25 metre high tree after 1.5 seconds. The tree is 60 metres away from the point from which the ball was hit. Assume $g = 10 \text{ m s}^{-2}$.



- (i) Show that the horizontal and vertical displacement of the ball are given by $x = Vt \cos \theta$ and $y = -5t^2 + Vt \sin \theta$. 2

- (ii) Find the angle of projection of the ball to the nearest minute. 2
- (iii) Find the initial speed of the ball. 1

5. (Riverview 2020 Q13a) A particle is projected from a point O on level horizontal ground with a speed of 21 m s^{-1} at an angle θ to the horizontal. At time T seconds, the particle passes through the point $B(12, 2)$.

Neglecting the effects of air resistance, the equations describing the motion of the particle are:

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2$$

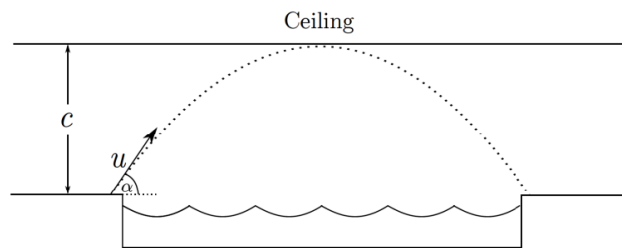
where t is the time in seconds after projection, $g \text{ m s}^{-2}$ is the acceleration due to gravity where $g = 9.8 \text{ m s}^{-2}$ and x and y are measured in metres. Do NOT prove these equations.

- (i) By considering the horizontal component of the particle's motion, show that 1

$$T = \frac{4}{7} \sec \theta$$

- (ii) By considering the vertical component of the particle's motion and, using the result from part (a) (i), show that $4 \tan^2 \theta - 30 \tan \theta + 9 = 0$. 2
- (iii) Find the particle's least possible flight time from O to B . Give your answer correct to two decimal places. 2

6. (Ascham 2023 Q14d) Queenie brings a football to a swimming pool and wishes to kick it to the other side without it landing in the water or bouncing off the ceiling. The ceiling is c metres high and the football will be kicked with an initial speed of $u \text{ m/s}$ and angle of elevation of α .



The equations of motion of the football are given by, using $g = 10 \text{ m/s}^2$:

$$\dot{x} = u \cos \alpha \quad \dot{y} = -10t + u \sin \alpha$$

$$x = ut \cos \alpha \quad y = -5t^2 + ut \sin \alpha$$

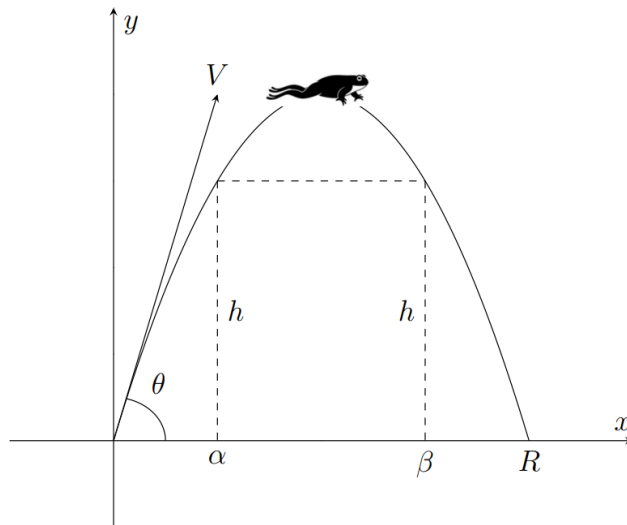
(Do not prove these)

- i) In order to improve the ball's chances of landing on the other side, Queenie kicks the ball so that it just touches the ceiling at its highest point in the trajectory.

Show that the speed required is: $u = \frac{2\sqrt{5c}}{\sin \alpha} \text{ m/s}$. 2

- ii) The length of the pool is $4\sqrt{3}$ times the height of the ceiling. Find the steepest angle at which Queenie can successfully kick the ball to the other side. 3

7. (Sydney Girls 2025 Q14a) A frog leaps with an initial speed of $V \text{ ms}^{-1}$.
 The angle of projection is θ , as shown below.
 The horizontal range of the frog is R .
 Let α and β be the x -coordinates of the frog when $y = h$.



The vector equation of motion is given by

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}. \quad [\text{Do not prove this.}]$$

(i) Show that $R = \frac{2V^2 \sin \theta \cos \theta}{g}$. 1

(ii) Hence, show that $y = x \left(1 - \frac{x}{R}\right) \tan \theta$. 2

(iii) Hence, show that α and β are the roots of 1

$$x^2 \tan \theta - xR \tan \theta + Rh = 0.$$

(iv) Deduce that $(\beta - \alpha)^2 = R(R - 4h \cot \theta)$. 2

8. (Barker 2025 Q14a) A stone is thrown from the top of a 60 m cliff at an angle of 25° below the horizontal. It lands in the ocean 45 m from the base of the cliff. Assume $g = 10 \text{ ms}^{-2}$.

(i) Taking the origin as the point where the stone is thrown and starting with the equations $\ddot{y} = -g$, and $\ddot{x} = 0$, show that the path of the stone can be represented by $y = -x \tan 25^\circ - \frac{5x^2}{V^2 \cos^2 25^\circ}$, where V is the initial velocity. 3

(ii) Hence find the initial velocity of the stone, to two decimal places. 1

(iii) Hence find the angle the stone enters the ocean, to the nearest degree. 2

9. (Sydney Boys 2020 Q14a) A golfer hits a golf ball from a point O with speed $V \text{ m/s}$ at an angle θ above the horizontal, where $0^\circ < \theta < 90^\circ$. The ball just passes over a 2.25 m high tree after 1.5 seconds. The tree is 60 metres away on level ground from the point from which the ball was hit. The acceleration due to gravity is 10 m/s^2 .

The position vector, $\underline{r}$, at any time, t seconds, is given by

$$\underline{r}(t) = Vt \cos \theta \underline{i} + (Vt \sin \theta - 5t^2) \underline{j} \quad (\text{Do NOT prove})$$

- (i) What is the initial angle, θ , of projection of the golf ball, to the nearest minute? **3**
- (ii) What is the initial speed (V m/s) of the golf ball? **1**
- (iii) How far from the base of the tree does the ball strike the ground? Leave your answer correct to the nearest metre. **1**